Problem: Determine a one-parameter Ritz solution for an isotropic rectangular plate with edges $x = 0, a$ clamped and edges $y = 0, b$ simply supported, and subjected to a transverse center point load P_0. Use the approximated solution in Eq. (1). Determine the maximum deflection of the rectangular plate. The thickness, modulus of elasticity and Poisson ratio of the plate are h, E and ν, respectively.

$$w(x, y) = c\left(1 - \cos\left(\frac{2\pi x}{a}\right)\right)\sin\left(\frac{\pi y}{b}\right) \tag{1}$$

Solution

A one-parameter function is given as

$$w(x, y) = c\left(1 - \cos\frac{2\pi x}{a}\right)\sin\frac{\pi y}{b}$$

First, we should calculate derivatives of the function to check essential boundary conditons.

So, we have clamped and simply supported boundary conditions;

– Clamped at $x = 0, a$: $w = 0$ and $\dfrac{\partial w}{\partial x} = 0$,

– Simply supported at $y = 0, b$: $w = 0$, All essential conditions are satisfied. Now we want to find the total potential energy;

$$\Pi = \frac{D}{2}\int_0^a\int_0^b\left[(w_{xx}+w_{yy})^2-2(1-\nu)(w_{xx}w_{yy}-w_{xy}^2)\right]dx\,dy-P_0w(\frac{a}{2},\frac{b}{2}) \tag{2}$$

Substituting the obtained derivatives into Eq. (2) gives

The work done by the point load is

Integrating the expressions, we obtain

Simplifying the total potential energy,

$$\boxed{c = \frac{P_0}{D\pi^4\left(\dfrac{2b}{a^3} + \dfrac{1}{ab} + \dfrac{3a}{8b^3}\right)}}$$

$$\boxed{w_{max} = \frac{24P_0(1-\nu^2)}{\pi^4 Eh^3\left(\dfrac{2b}{a^3} + \dfrac{1}{ab} + \dfrac{3a}{8b^3}\right)}}$$